

Image Captioning on Flickr8k

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1 Introduction

We design, train and compare neural image-captioning systems on the Flickr8k dataset. Given an image, each model generates a natural-language description. We implement the two required architectures: the **Show and Tell** encoder-decoder [1] and the attention-based **Show, Attend and Tell** [2], under a single, shared data, training and evaluation pipeline. On top of the requirement of two optional extensions we address all five: beam search, attention visualization, an alternative (dot-product) attention mechanism, a Transformer decoder, and pretrained GloVe embeddings. Crucially, every decoder uses the *same* fine-tuned ResNet-18 encoder, so performance differences are attributable to the decoder. The full implementation is in the accompanying Jupyter notebook, and all runs were tracked with Weights & Biases.

2 Data and preprocessing

Flickr8k contains 8,091 images, each annotated with five human captions. We use the provided split unchanged: **6,473 training images** (32,365 image-caption pairs, since every caption is a separate training target) and **1,618 test images**. The word-level vocabulary is built from the training captions only (no test leakage), keeping tokens that occur at least 5 times, which yields **2,649 tokens** including the special tokens <pad>, <bos>, <eos> and <unk>. Captions are lower-cased, tokenized, wrapped with begin/end markers, and padded to length 40.

Images are resized to 224×224 and normalized with ImageNet statistics. Because the dataset is small, the training images are augmented on the fly, random-resized crop (scale 0.65–1.0), horizontal flip, colour jitter (brightness/contrast/saturation 0.3, hue 0.1) and occasional grayscale, while the test transform is deterministic (resize-256, centre-crop-224). Decoded images are cached once to avoid redundant JPEG decoding, with augmentation re-applied live each epoch so regularization is unaffected.

3 Model architectures

Shared encoder. All models use a torchvision ResNet-18 pretrained on ImageNet with its classification head removed. The baseline takes the global average-pooled vector and projects it to 512-d (followed by batch-norm); the attention and Transformer models instead keep the final $512 \times 7 \times 7$ convolutional feature map, reshaped to **49 spatial locations of 512-d each**. The encoder is fine-tuned (unfrozen) at a low learning rate.

Show and Tell (baseline). The 512-d image vector initializes the hidden and cell state of an LSTM decoder (LSTMCell, hidden size 512, word embedding 256). At each step the previous word embedding is fed to the cell and mapped to vocabulary logits. Training is teacher-forced through a `forward()` method. Inference uses greedy decoding in a separate `generate()` method.

Show, Attend and Tell. At every step, additive (Bahdanau) attention scores the 49 locations, $e_i = v^\top \tanh(W_e a_i + W_d h_t)$, and forms a context vector as the softmax-weighted sum of encoder features; a doubly-stochastic gate $\beta = \sigma(W_h)$ scales the context, which is then concatenated with the word embedding as LSTM input (attention dim 256). A `generate_with_attention()` method returns the per-step attention maps used in Section 7.

4 Experimental setup

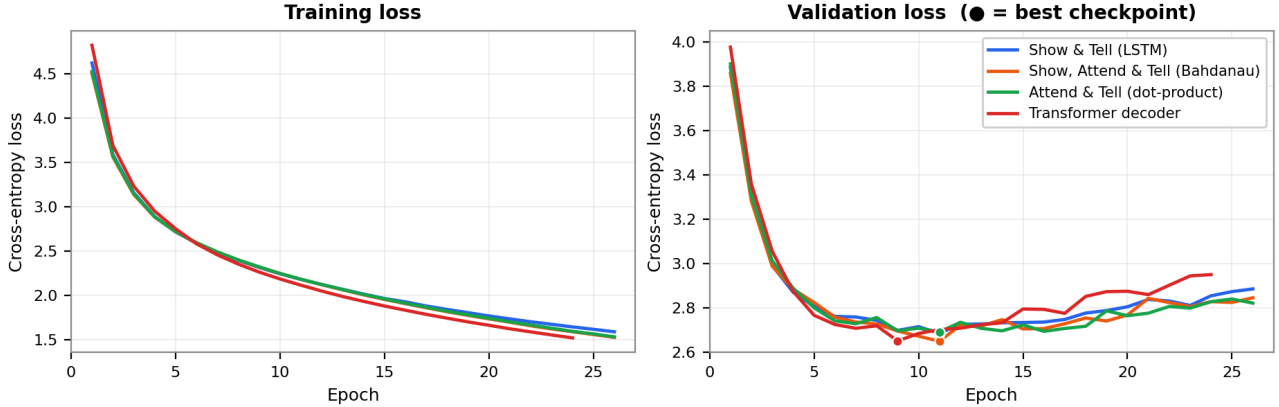
For a fair comparison all models share one training recipe: AdamW with decoder learning rate 5×10^{-4} and encoder learning rate 1×10^{-5} , weight decay 1×10^{-3} , a 3-epoch linear warmup followed by cosine decay to 10% of the peak, gradient-norm clipping at 5.0, dropout 0.2, batch size 64, and bf16 mixed precision on a single GPU. We train up to 100 epochs with early stopping (patience 15) and reload the best-validation checkpoint before reporting. The objective is token-level cross-entropy with padding masked (the <pad> token is ignored). The Transformer uses learning rate of 3×10^{-4} , weight decay 1×10^{-4} , a 5-epoch warmup and dropout 0.1. Evaluation reports (i) **teacher-forced perplexity** (the exponential of the mean per-token cross-entropy over the test set) and (ii) **corpus BLEU-1 to BLEU-4** computed with NLTK (smoothing method 1), comparing greedy generations against all five reference captions per image.

5 Quantitative results

Table 1 reports test perplexity, corpus BLEU-1 to BLEU-4, parameter count and the best-checkpoint epoch for the four decoders (all sharing the same fine-tuned encoder). Figure 1 shows the loss curves. Training loss decreases monotonically for every model, whereas validation loss bottoms out around epochs 9–11 and then climbs, very obvious overfitting, expected with only ~6.5k training images despite augmentation and regularization. The three LSTM decoders are very close: attention lifts BLEU-4 over the baseline by ~0.008 and the dot-product variant has the best LSTM perplexity. The **Transformer decoder wins on every metric** (BLEU-4 0.205, +0.027 over the baseline) while using *fewer* parameters and reaching its best checkpoint earlier (epoch 9). Since the encoder is identical across models, this improvement comes from the decoder: self-attention over generated tokens plus multi-head cross-attention to all 49 patches captures sequence structure better than a single LSTM cell can.

Table 1. Test-set results (best checkpoint, greedy decoding). Best per column in bold.

Decoder (shared ResNet-18 enc.)	Perplexity	BLEU-1	BLEU-2	BLEU-3	BLEU-4	Params	Best epoch
Show and Tell (LSTM) [1]	14.61	0.5871	0.4068	0.2708	0.1782	15.58 M	11
Show, Attend and Tell (Bahdanau) [2]	14.68	0.5883	0.4096	0.2774	0.1862	16.89 M	11
Attend and Tell (dot-product)	14.27	0.5915	0.4148	0.2796	0.1866	16.89 M	11
Transformer decoder	14.21	0.6151	0.4340	0.2983	0.2047	15.84 M	9

**Figure 1.** Training (left) and validation (right) cross-entropy loss for the four decoders. Dots mark the best-validation checkpoint that is reloaded for evaluation.

6 Extensions

Beam search. For Show, Attend and Tell, length-normalized beam search (beam size 5) improves all BLEU scores over greedy decoding on a fixed 500-image subset (Table 2), with the largest gain at BLEU-4 (+0.026). This is consistent with beam search helping most where sentence-level structure matters, by keeping several hypotheses open instead of committing to a locally optimal token. Our implementation decodes one image at a time and is therefore much slower, so it was evaluated on a subset.

Table 2. Greedy vs. beam search.

Decoding (Show, Attend and Tell, 500 imgs)	BLEU-1	BLEU-2	BLEU-3	BLEU-4
Greedy	0.6005	0.4233	0.2874	0.1912
Beam ($k = 5$)	0.6081	0.4387	0.3113	0.2172

Alternative attention. Replacing additive (Bahdanau) attention with scaled dot-product attention, everything else fixed, gives near-identical results (Table 1, rows 2–3) and slightly better perplexity but essentially equal BLEU and overlapping loss curves. With only 49 locations and attention dim 256 the two scoring functions have similar effective capacity, so the multiplicative interaction brings no clear advantage here.

Pretrained embeddings (GloVe). We compare random vs. GloVe-6B-100d initialization for a 100-d Show and Tell (embeddings fine-tuned, not frozen). Its vocabulary coverage is **99.5%** (2,635/2,649). The two are statistically indistinguishable (Table 3). GloVe gives marginally lower perplexity but slightly lower BLEU-1, and the curves overlap. Because embeddings are fine-tuned, the random model learns adequate word vectors within a few epochs. With near-total coverage of common words and a grounding-limited task, GloVe’s prior adds little. Both 100-d variants trail the 256-d models, suggesting the narrower embedding costs some capacity.

Table 3. GloVe vs. random embedding initialization (better value per metric in bold).

Embedding init (Show and Tell, 100-d)	Perplexity	BLEU-1	BLEU-2	BLEU-3	BLEU-4
Random	15.43	0.6061	0.4204	0.2802	0.1829
GloVe-6B-100d	15.32	0.5937	0.4137	0.2777	0.1824

7 Qualitative analysis



Figure 2. Show, Attend and Tell: attention maps overlaid on a test image for each generated word. The model attends to the child’s body for person words but still hallucinates “a man and a woman” for a single silhouette; function words (“a”, “and”) are diffuse.

Attention. Content words attend to plausible regions, e.g. person words concentrate on the figure, “standing” on the body, and (lower row, not shown) “lake” on the water. Function words, on the other side are spread out. However, correct attention does not guarantee a correct word. The model looks at the right region yet outputs “a man and a woman” for one child, defaulting to the most frequent training concept for an ambiguous shape. The 7×7 (49-location) grid is also coarse for fine localization.

Captions. Table 4 shows representative greedy captions from the best (Transformer) model. It produces fluent, well-formed sentences and handles common scenes (“two dogs are playing with a toy”), but the recurring failures are overly generic templates, hallucinated objects, wrong counts, and occasional degenerate repetition. These errors persist across all four decoders, indicating the bottleneck is **visual grounding** (recognition), not language modelling.

Table 4. Representative successful and failure cases.

Generated (Transformer, greedy)	Reference (one of five)	Error type
two dogs are playing with a toy	A brown and black lab. . . the black lab is catching a toy in its mouth.	— correct
a man is standing in a lake with a in the background	Two people are at the edge of a lake, facing the water and the city skyline.	generic + repetition
a boy in a red shirt is holding a red and white bat	A person in blue and red ice climbing with two picks.	hallucination / wrong activity
a man is standing on a hill with a mountain in the background	Three people overlook a green valley.	wrong count + generic
a little boy is sitting on a bench with a baby in the background	A young boy overlooking a crowd of inflatable boats.	hallucinated objects

On the metrics. The steep BLEU-1 to BLEU-4 drop shows the models get individual words right but struggle to assemble longer correct phrases. BLEU-1 is inflated by matching against five references and by frequent common words, and BLEU does not penalize hallucinations whose neighbours still overlap a reference. Perplexity is teacher-forced, so it measures language-model fit rather than free-generation quality. This is why it occasionally disagrees with BLEU (e.g. the attention LSTM has slightly worse perplexity than the baseline despite a lower validation loss).

8 Limitations and possible improvements

The dominant limitation is **data size**: $\sim 6.5k$ training images cause early overfitting that augmentation, dropout and weight decay only slow. The second is the **encoder**: ResNet-18 caps visual grounding, and most errors are recognition errors rather than language errors. Third, results come from **single runs**, so the ~ 0.01 BLEU-4 gaps among the LSTM variants are within run-to-run noise. Only the Transformer’s ~ 0.027 gain is clearly meaningful. Promising directions include a stronger encoder (ResNet-50/101, ViT or CLIP features), metrics that correlate better with human judgement (CIDEr, METEOR, SPICE), label smoothing and scheduled sampling to reduce exposure bias, batched beam search, training across multiple seeds with significance testing, and CIDEr-optimized fine-tuning (self-critical sequence training).

[1] O. Vinyals, A. Toshev, S. Bengio, D. Erhan. *Show and Tell: A Neural Image Caption Generator*. CVPR 2015.

[2] K. Xu et al. *Show, Attend and Tell: Neural Image Caption Generation with Visual Attention*. ICML 2015.